RAM CHARAN VANGA
RA2511027010164
Sub ject : DSA

Unit 3, S7, SLO1 - Implementation of Stack using Array

Build the code for Implementation of Stacks using Array:
a. Declare the stack with size MAX, initialize 'top'
b. Write a function push with arguments int st[ ] and int val
c. Write a function pop with argument int st[ ]
d. Write a function peek with argument int st[ ]

Answer: /* a. Declaration and Initialization */

```
#define MAX 100
int st[MAX];
int top = -1;

/* b. Push Function */
void push(int st[], int val) {
if (top == MAX - 1) {
printf("
Stack Overflow! Cannot push %d\n", val);
return;
}
top++;
st[top] = val;
printf("
Pushed %d to stack\n", val);
}

/* c. Pop Function */
int pop(int st[]) {
if (top == -1) {
printf("
Stack Underflow! Stack is empty\n");
return -1;
}
int val = st[top];
top--;
return val;
}

/* d. Peek Function */
int peek(int st[]) {
if (top == -1) {
printf("
Stack is Empty!\n");
return -1;
}
return st[top];
```

```
}
```